

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 42

CLIMATE CHANGE - 3

4 SEA LEVEL WILL RISE 1-4 FEET BY 2100

- Global sea level has risen by about 8 inches since record keeping began in 1880. It is projected to rise another 1 to 4 feet by 2100.
- This is the result of added water from melting land ice and the expansion of seawater as it warms.
- In the future, storm surges and high tides could combine with sea level rise to increase flooding in many regions.
- Ocean waters will continue to warm and sea level will continue to rise for many centuries because the oceans take a very long time to respond to warmer conditions at the Earth's surface.



5 CLIMATE CHANGE AND WILDLIFE

- Global warming is likely to be the greatest cause of species extinctions this century.
- The IPCC says a 1.5°C average rise may put 20-30% of species at risk of extinction.
- If the planet warms by more than 3°C , most ecosystems will struggle.
- Climate Change is happening too quickly for many species to adapt.



5 CLIMATE CHANGE AND WILDLIFE

SNOW LEOPARDS

- Warming in the Himalayas is happening very rapidly. This is prime snow leopard habitat.
- Continued warming will cause their range to shrink as the tree-line moves higher up the mountains.
- This will not only fragment and isolate snow leopard populations, but it will severely affect their prey too.



ASIAN RHINOS

- The Indian one-horned rhinos live on floodplain grasslands in northern India and Nepal.
- They rely on the annual monsoon to bring sufficient and timely rain, to replenish the vegetation they feed on.
- But a changing climate could disrupt this seasonal pattern and bring regular droughts or floods.





AFRICAN ELEPHANTS

- In Africa, changes in rainfall will either bring too much rain – causing floods – or too little rain – bringing more drought and wildfires.
- These changes may cause some areas to simply become unsuitable for certain species to live in.
- African elephants can drink up to 225 liters of water each day, so changing weather patterns may mean they have to travel further in search of water – moving outside protected areas and coming into more contact with people.





POLAR BEARS

The Arctic is warming roughly twice as fast as the global average, causing the ice that polar bears depend on to melt away.



IMPACTS OF CLIMATE CHANGE

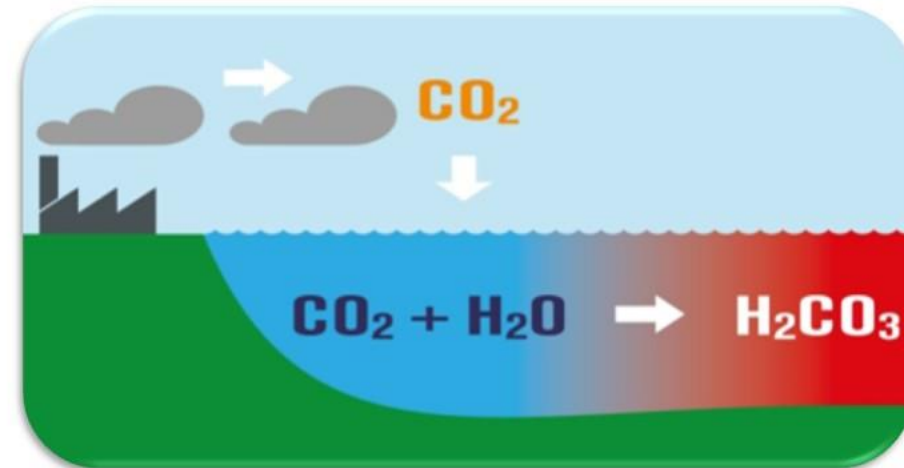
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CLIMATE CHANGE AND OCEANS

- Oceans are vital carbon sinks, they absorb huge amounts of carbon dioxide.
- Increased water temperatures and higher carbon dioxide concentrations than normal, which make oceans more acidic, are already having an adverse impact on oceans.

OCEAN ACIDIFICATION

- It is the ongoing decrease in the pH of oceans caused by absorption of excess carbon dioxide from the atmosphere.
- Ocean Acidification is also known as the 'Evil Twin of Global Warming' and 'The Other CO₂ Problem'.



CAUSES OF OCEAN ACIDIFICATION

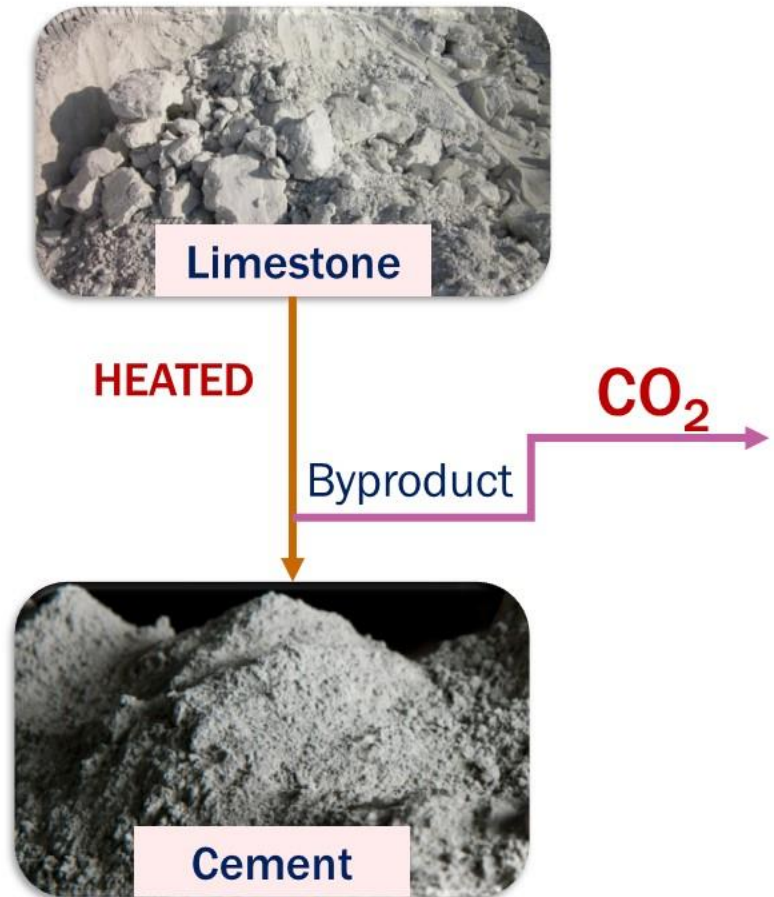
- Oceans are the largest carbon sink. Around 30-40% of the CO_2 released from human activities into the atmosphere is dissolved into oceans, rivers and lakes.
- CO_2 reacts with water to form carbonic acid. So, all the activities that cause release of CO_2 into the atmosphere cause ocean acidification.
- Such activities are:
 - **Burning fossil fuels**
 - ✓ Burning fossil fuels emits CO_2 . Coal based thermal power plants are a major source of CO_2 emissions in India. Coal releases twice as much as CO_2 as petroleum.



CAUSES OF OCEAN ACIDIFICATION

➤ Industries

- ✓ Many industrial processes emit carbon dioxide directly through fossil fuel combustion.
- ✓ There are four main types of industrial processes that are a significant source of carbon dioxide emissions.
- ✓ Cement production produces the most amount of carbon dioxide amongst all industrial processes.
- ✓ To create the main ingredient in cement, calcium oxide, limestone is chemically transformed by heating it to very high temperatures.
- ✓ This process produces large quantities of carbon dioxide as a byproduct of the chemical reaction.



CAUSES OF OCEAN ACIDIFICATION

➤ Deforestation

- ✓ Trees absorb CO_2 from the air. Loss of forestlands accounts for 20 percent of global CO_2 emissions. Burning trees for fuel and natural decomposition of forest material also releases CO_2 .



➤ Coastal Acidification

- ✓ Human activities also contribute to acidification in coastal waters. When compounds such as Sulfur Dioxide and Nitrogen Oxides are released, they react with water vapour in the atmosphere to form acid rain.



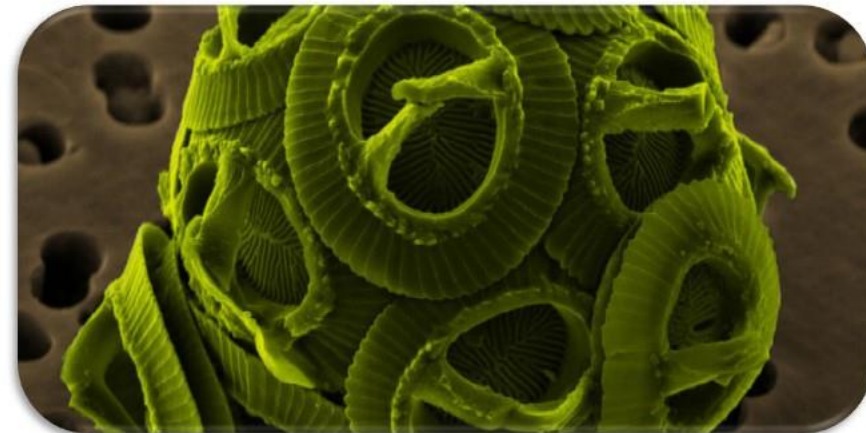
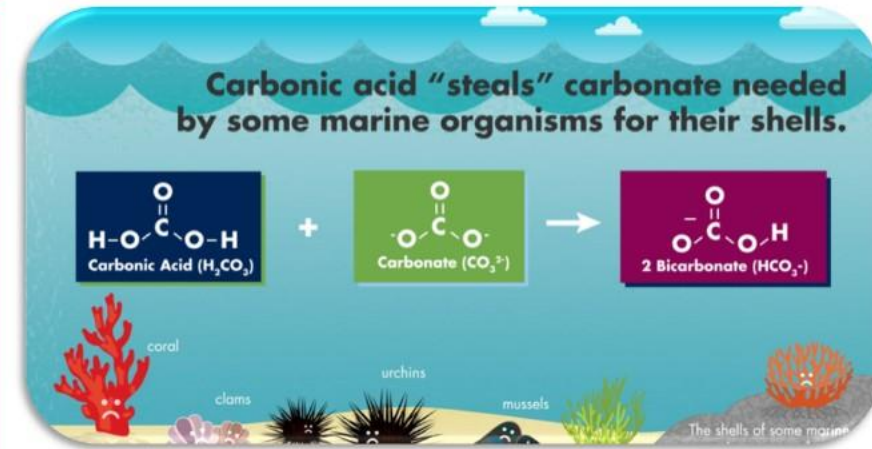


EFFECTS OF OCEAN ACIDIFICATION

- ✓ If CO_2 is absorbed normally by the oceans, then there is abundance of carbonate ions, which are essential for the shell formation of many aquatic organisms.
- ✓ Calcium Carbonate is the building block for the skeletons and shells of many marine organisms like corals, molluscs (Oysters) and crustaceans (shrimp, crab, lobsters etc.)
- ✓ If CO_2 is absorbed normally by oceans then there is abundance of carbonate ions which are essential for calcium carbonate shell formation.
- ✓ But due to excess absorption of CO_2 bicarbonate ions are produced. So, carbonate ions are less available and calcification is harder to achieve for many marine organisms.



- ✓ Due to acidification the existing shells of organisms are vulnerable to dissolution.
- ✓ Increasing ocean acidity has been shown to significantly reduce the reef building ability of corals and also cause coral bleaching.
- ✓ Increasing acidity decreases the metabolic rates and weakens the immune system of aquatic organisms.
- ✓ Biodiversity of the oceans and commercial fisheries are threatened because acidification harms calcifying organisms, which are important part of the oceanic food webs like Calcerous Phytoplankton.



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